

Exe-3

Q1) Find the rank of matrix by normal form.

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ 1 & -2 & 6 & -7 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ 1 & -2 & 6 & -7 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, \quad R_3 \rightarrow R_3 - R_1, \quad \& \quad R_4 \rightarrow R_4 + R_1$$

$$A \sim \begin{bmatrix} 1 & 2 & -1 & 4 \\ 0 & 0 & 5 & -4 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 5 & -3 \end{bmatrix}$$

$$C_2 \rightarrow C_2 - 2C_1, \quad C_3 \rightarrow C_3 + C_1$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -4 & 5 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 5 & -3 \end{bmatrix}$$

$$\& \quad C_4 \rightarrow C_4 - 2C_1$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 5 & -4 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 5 & -3 \end{bmatrix}$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -4 & 5 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 3 & 5 & 0 \end{bmatrix} \quad C_2 \leftrightarrow C_4$$

$$R_2 \rightarrow -R_2$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -5/4 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 3 & 5 & 0 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - 3R_2$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -5/4 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 5/4 & 0 \end{bmatrix}$$

$$R_3 \rightarrow \frac{R_3}{4}$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -5/4 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 5/4 & 0 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - 5/4 R_3$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -5/4 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - \frac{5}{4} R_3$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -5/4 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + \frac{5}{4} R_3$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rho(A) = (3)$$

Q2) Find the value of λ and u the system of linear eq?

$$x + y + z = 6$$

$$x + 2y + 5z = 10$$

$$2x + 3y + \lambda z = u + v$$

① linear give soln

② no soln

③ infinite soln and also find the value

$$\lambda = 2 \text{ and } u = 7$$

$$AX = B$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 5 \\ 2 & 3 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ u \end{bmatrix}$$

$$C \sim [A:B]$$

$$C \sim \begin{bmatrix} 1 & 1 & 1 & : & b \\ 1 & 2 & 5 & : & 10 \\ 2 & 3 & \lambda & : & u \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - 2R_1$$

$$C \sim \begin{bmatrix} 1 & 1 & 1 & : & b \\ 0 & 1 & 4 & : & 4 \\ 0 & 1 & \lambda-2 & : & u-2b \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$C \sim \begin{bmatrix} 1 & 1 & 1 & : & b \\ 0 & 1 & 4 & : & 4 \\ 0 & 0 & \lambda-6 & : & u-16 \end{bmatrix}$$

① Unique Solⁿ

$$\text{if } \lambda - 6 \neq 0$$

$$\lambda \neq 6$$

② No Solⁿ

$$\text{if } \lambda - 6 = 0$$

$$\lambda = 6$$

$$u - 16 \neq 0$$

$$u \neq 16$$

iii) Infinite solⁿ

$$\lambda - b = 0$$

$$\lambda = b$$

$$\lambda - 1b = 0$$

$$\lambda = 1b$$

Given, $\lambda = 2$ and $\mu = 8$

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 4 & : & 4 \\ 0 & 0 & -4 & : & -8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 4 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 8 \end{bmatrix}$$

$$x + y + z = 6 \quad \text{--- (i)}$$

$$0x + y + 4z = 4 \quad \text{--- (ii)}$$

$$0x + 0y - 4z = -8 \quad \text{--- (iii)}$$

from (iii)

$$\boxed{z = 2}$$

Put $z = 2$ in eqⁿ (ii)

$$\boxed{y = -4}$$

Put $y = -4$ in eqⁿ (i) and $z = 2$ in eqⁿ (i)

$$x + 4y + z = 6$$

$$x - 4 + 2 = 6$$

$$x - 2 = 6$$

$$x = 8$$

Q3 Find the eigen value and eigen vector of

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

→ Characteristic eqⁿ

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)[(2-\lambda)^2 - 1] + 1[\lambda - 2 + 1] + 1[\lambda - 1] = 0$$

$$(2-\lambda)[\lambda^2 - 4\lambda + 3] + [\lambda - 1] + [\lambda - 1] = 0$$

$$-\lambda^3 + \lambda^2[4+2] + \lambda[-3-8+2] + [6-2] = 0$$

$$-\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0$$

$$\text{If we let } \lambda = 1 \Rightarrow 1 - 6 + 9 - 4 = 0$$

$$\lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$

$$\lambda^3 - \lambda^2 - 5\lambda^2 + 5\lambda + 4\lambda - 4 = 0$$

$$\lambda^2(\lambda - 1) - 5\lambda(\lambda - 1) + 4(\lambda - 1) = 0$$

$$(\lambda - 1)(\lambda^2 - 5\lambda + 4) = 0$$

$$(\lambda - 1)(\lambda^2 - 4\lambda - \lambda + 4) = 0$$

$$(\lambda - 1)(\lambda(\lambda - 4) - 1(\lambda - 4))$$

$$(\lambda - 1)(\lambda - 1)(\lambda - 4)$$

vector

$\lambda = 1, 1, 4$ (Eigen value)

let $\lambda = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ be the Eigen vector for $\lambda = 4$

$$(A - \lambda I)X_1 = 0$$

$$\begin{bmatrix} -2 & 1 & 1 \\ -1 & -2 & 1 \\ 1 & -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$-2x - y + z = 0$$

$$-x - 2y - z = 0$$

$$\frac{x}{1-(-1)} = \frac{-y}{2-(-1)} = \frac{z}{4-1}$$

$$\frac{x}{3} = \frac{y}{-3} = \frac{z}{3}$$

$$\frac{x}{1} = \frac{y}{-1} = \frac{z}{1} = 3k$$

$$X_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

let $\lambda_2 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ be the Eigen vector for $\lambda = 1$

$$(A - \lambda I)X_2 = 0$$

$$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$R_2 \rightarrow R_2 + R_1, R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$x - y + z = 0$$

$$\text{let } y = k_1, z = k_2$$

$$x - y + z = 0$$

$$x - k_1 + k_2 = 0$$

$$x = k_1 - k_2$$

$$\begin{bmatrix} k_1 - k_2 \\ k_1 \\ k_2 \end{bmatrix}$$

$$k_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$k_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Eigen vectors :-

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Tute - II

Q1 $A = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$ find A^{-1}

$$|A| = 1(-1-3) - 2(-1-1) - 2(3-1)$$

$$-4 + 4 - 4 = -4$$

co-factor of matrix $A = \begin{bmatrix} -4 & 2 & 2 \\ 4 & -1 & 1 \\ 4 & -3 & -1 \end{bmatrix}$

~~$\text{Adj } A = \begin{bmatrix} -4 & 4 & 4 \\ 2 & -1 & 3 \\ 2 & 1 & -1 \end{bmatrix}$~~

now

$$A^{-1} = \frac{1}{|A|} \text{Adj } A$$

$$= \frac{1}{-4} \begin{bmatrix} -4 & 4 & 4 \\ 2 & -1 & -3 \\ 2 & 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & -1 \\ -\frac{1}{2} & \frac{1}{4} & \frac{3}{4} \\ -\frac{1}{2} & -\frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

Q2) State Cayley-Hamilton theorem and

verify $A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

⇒ Cayley-Hamilton Theorem State that Every square matrix satisfies its own characteristic equation.

If $|A - \lambda I| = (-1)^n (\lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \dots + a_n)$ be the characteristic polynomial of an $n \times n$ matrix $A = (a_{ij})$, then the matrix eqⁿ.

$$\lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \dots + a_n I = 0 \text{ is satisfied}$$

$$\text{by } \lambda = A \text{ i.e.,}$$

$$A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I = 0$$

now,

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 1 & 2 \\ 0 & -2-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(12-\lambda)(3-\lambda) - 1(6-0) + 2(0) = 0$$

$$(1-\lambda)(-6+2\lambda-3\lambda+12) = 0$$

$$(1-\lambda)(-6-\lambda+12) = 0$$

$$(1-\lambda)(\lambda^2-\lambda-6) = 0$$

$$\lambda^2-\lambda-6-\lambda^3+\lambda^2+6\lambda = 0$$

$$-\lambda^3+2\lambda^2+5\lambda-6 = 0$$

Now multiply by $\lambda^{-1} = \lambda^3 - 2\lambda^2 - 5\lambda + 6 = 0$

$$\lambda^2 - 2\lambda - 5 + 6\lambda^{-1} = 0$$

$$-A^2 + 2A + 5 - 6A^{-1} = 0$$

$$6A^{-1} = -A^2 + 2A + 5$$

$$6A^{-1} = \begin{bmatrix} 2 & 1 & -9 \\ 0 & -4 & 0 \\ 0 & 0 & -9 \end{bmatrix} + \begin{bmatrix} 2 & 2 & 4 \\ 0 & -4 & 0 \\ 0 & 0 & 6 \end{bmatrix} + \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 3 & -4 \\ 0 & -8 & 0 \\ 0 & 0 & -3 \end{bmatrix} + \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{6} \begin{bmatrix} 6 & 3 & -4 \\ 0 & -3 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} & -\frac{2}{3} \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix}$$

Q3) let $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ find the module matrix P.

$$\Rightarrow |A - \lambda I| = 0$$

$$\begin{bmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{bmatrix} = 0$$

$$(6-\lambda)(3-\lambda)(3-\lambda) - 1 + 2(-2(3-\lambda) + 2) + 2(2 - 2(3-\lambda)) = 0$$

$$(6-\lambda)(9 + \lambda^2 - 6\lambda - 1) + 2(-6 + 2\lambda + 2) + 2(2 - 6 + 2\lambda) = 0$$

$$(6-\lambda)(\lambda^2 - 6\lambda + 8) + 4\lambda - 8 + 8 + 2\lambda = 0$$

$$6\lambda^2 - 36\lambda + 48 - \lambda^3 + 6\lambda^2 - 8\lambda + 4\lambda - 16 + 4\lambda = 0$$

$$-\lambda^3 + 12\lambda^2 - 36\lambda + 32 = 0$$

$$\Rightarrow \lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

$$\Rightarrow \lambda^3 - 12\lambda^2$$

$$\lambda^2(\lambda - 12)$$

$$\Rightarrow (\lambda - 2)^2(\lambda - 8) = 0$$

$$\lambda = 2, 2, 8$$

Eigen vector for $\lambda = 2$

$$\begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{array}{l} R_1 \rightarrow 2R_1 \\ R_2 \rightarrow R_1 + R_2 \\ R_3 \rightarrow R_2 + R_3 \end{array}$$

$$\begin{bmatrix} 2 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$2x_1 - x_2 + x_3 = 0$$

this equation is satisfied by $x_1 = 0, x_2 = 1, x_3 = 1$

$$x_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

again, $x_1 = 1, x_2 = 3, x_3 = 1$

$$x_2 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

for $\lambda = 8$

$$\begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-2x_1 - 2x_2 + 2x_3 = 0$$

$$-2x_1 - 5x_2 - x_3 = 0$$

$$\frac{x_1}{12} = \frac{x_2}{-6} = \frac{x_3}{6} \Rightarrow \frac{x_1}{2} = \frac{x_2}{-1} = \frac{x_3}{1}$$

$$x_3 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\therefore P = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 3 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

Q4) Find the value diagonalization of matrix A =

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{bmatrix} -(2+\lambda) & 2 & -3 \\ 2 & (1-\lambda) & -6 \\ -1 & -2 & -\lambda \end{bmatrix} = 0$$

$$-(2+\lambda) [(1-\lambda)-\lambda-12] - 2[-2\lambda-6] - 3[-4+(1-\lambda)] = 0$$

$$-(2+\lambda) [-\lambda + \lambda^2 - 12] + 4\lambda + 12 + 12 - 3 + 3\lambda = 0$$

$$2\lambda - 2\lambda^2 + 24 + \lambda^2 - \lambda^3 + 12\lambda + 4\lambda + 12 + 12 - 3 + 3\lambda = 0$$

$$-\lambda^3 - \lambda^2 + 21\lambda + 45 = 0$$

$$\lambda^3 + \lambda^2 - 21\lambda - 45 = 0$$

$$\lambda^3 - 5\lambda^2 + 6\lambda^2 - 32\lambda + 9\lambda - 45 = 0$$

$$\lambda^2(\lambda - 5) + 6\lambda(\lambda - 5) + 9(\lambda - 5) = 0$$

$$(\lambda - 5)(\lambda^2 + 6\lambda + 9) = 0$$

$$(\lambda - 5)\lambda(\lambda + 3) + 3(\lambda + 3) = 0$$

$$\lambda = 5, -3, -3$$

Eigen vectors for $\lambda = 5$

$$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$-7x + 2y - 3z = 0$$

$$2x - 4y - 6z = 0$$

$$\frac{x}{-24} = \frac{y}{-48} = \frac{z}{24}$$

$$\frac{x}{-1} = \frac{y}{-2} = \frac{z}{1}$$

$$x_1 = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$$

For $\lambda = -3$

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \quad \begin{array}{l} R_1 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 + R_1 \end{array}$$

$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$x + 2y - 3z = 0$$

let $x = 2, y = -1, z = 0$

$$x_2 = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$$

let $x = 3, y = 0, z = 1$

$$x_3 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

$$\therefore P = \begin{bmatrix} -1 & 2 & 3 \\ -2 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$|A| = 8$$

$$\rightarrow R_2 - 7R_1$$

$$\rightarrow R_3 + R_1$$

$$P^{-1} = \frac{1}{8} \begin{bmatrix} -1 & -2 & 3 \\ 2 & -4 & -6 \\ 1 & 2 & 5 \end{bmatrix}$$

$$D = P^{-1}AP$$

$$= \frac{1}{8} \begin{bmatrix} -1 & -2 & 3 \\ 2 & -4 & -6 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 6 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ -2 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \frac{1}{8} \begin{bmatrix} -5 & -10 & 15 \\ -4 & 12 & 18 \\ -3 & -6 & -15 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ -2 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \frac{1}{8} \begin{bmatrix} 40 & 0 & 0 \\ 0 & -24 & 0 \\ 0 & 0 & -24 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

Q5) A system of eqⁿ

$$2x - y = 3$$

$$x - 3y = 4$$

$$x + 2y = 1$$

Which solⁿ is this (i) unique solⁿ (ii) infinite solⁿ (iii) inconsistent

$$A = \begin{bmatrix} 2 & -1 \\ 1 & -3 \\ 1 & 2 \end{bmatrix} \Rightarrow B = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$$

$$C = [A : B]$$

$$\begin{bmatrix} 2 & -1 & : & 3 \\ 1 & -3 & : & 4 \\ 1 & 2 & : & 1 \end{bmatrix} \quad R_1 \rightarrow R_1 - R_3$$

$$\begin{bmatrix} 1 & -3 & : & 2 \\ 1 & -3 & : & 4 \\ 1 & 2 & : & 1 \end{bmatrix} \quad R_2 \rightarrow R_2 - R_3$$

$$\begin{bmatrix} 1 & -3 & : & 2 \\ 0 & -5 & : & 3 \\ 1 & 2 & : & 1 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & -3 & : & 2 \\ 0 & -5 & : & 3 \\ 0 & 5 & : & -1 \end{bmatrix} \quad R_3 \rightarrow R_3 + R_2$$

$$\begin{bmatrix} 1 & -3 & : & 2 \\ 0 & -5 & : & 3 \\ 0 & 0 & : & 2 \end{bmatrix}$$

$$r(A) = 2 \neq r(C) = n \quad \text{no of unknowns}$$

then it is inconsistent and
it has no solⁿ.

Tute - III

Q1) If $u = x^2 + y^2$ find the value of $\frac{\partial^2 u}{\partial x \partial y}$.

$$\frac{\partial u}{\partial y} = 2y$$

$$\frac{\partial^2 u}{\partial x \partial y} = 0$$

Q2) ~~Q2) State and prove Euler's theorem~~

~~If z is a homogeneous function of x, y of order n , then~~

~~$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = nz$$~~

Proof:- $z = x^n \cdot f\left(\frac{y}{x}\right)$

$$\frac{\partial z}{\partial x} = n \cdot x^{n-1} f\left(\frac{y}{x}\right) + x^n f'\left(\frac{y}{x}\right) \cdot \left(-\frac{y}{x^2}\right) z$$

$$x \frac{\partial z}{\partial x} = n x^n f\left(\frac{y}{x}\right) - x^{n-1} y f'\left(\frac{y}{x}\right) z$$

$$x \frac{\partial z}{\partial x} = n x^n f\left(\frac{y}{x}\right) - y x^{n-1} f'\left(\frac{y}{x}\right) - 0$$

$$\frac{\partial z}{\partial y} = x^n f'\left(\frac{y}{x}\right) \frac{1}{x}$$

$$\frac{\partial z}{\partial y} = x^{n-1} f'\left(\frac{y}{x}\right)$$

$$y \frac{\partial z}{\partial y} = y x^{n-1} f'\left(\frac{y}{x}\right) \quad \text{--- (1)}$$

0 + 03

$$\begin{aligned} x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} &= n x^n f\left(\frac{y}{x}\right) - y x^{n-1} f'\left(\frac{y}{x}\right) \\ &\quad + y x^{n-1} f'\left(\frac{y}{x}\right) \\ &= n x^n f\left(\frac{y}{x}\right) \end{aligned}$$

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = n z$$

Q3) If $z = \frac{x^3+y^3}{xy}$, find the value of $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y}$

$$\frac{\partial z}{\partial x} = \frac{3xyx^2 - (x^3+y^3)y}{(xy)^2} = \frac{3x^3y - x^3y - y^4}{(xy)^2}$$

$$= \frac{2x^3y - y^4}{(xy)^2}$$

$$x \frac{\partial z}{\partial x} = \frac{2x^4y - xy^4}{(xy)^2} \quad \text{--- (i)}$$

$$\frac{\partial z}{\partial y} = \frac{3yx^2 - (x^3+y^3)x}{(xy)^2}$$

$$= \frac{3y^3x - x^4 - y^3x}{(xy)^2} = \frac{2y^3x - x^4}{(xy)^2}$$

$$y \frac{\partial z}{\partial y} = \frac{2y^4x - x^4y}{(xy)^2} \quad \text{--- (ii)}$$

Add (i) and (ii)

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{2x^4y - xy^4}{(xy)^2} + \frac{2y^4x - x^4y}{(xy)^2}$$

$$= \frac{x^4y + y^4x}{(xy)^2} = \left(\frac{x^3+y^3}{(xy)^2} \right)$$

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{x^3+y^3}{xy}$$

Q4) 1) $u = \log(x^3 + y^3 + z^3 - 3xyz)$ Show that
 $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) u = \frac{-1}{(x+y+z)^2}$

$$\frac{\partial}{\partial x} = \frac{1}{(x^3 + y^3 + z^3 - 3xyz)} \times 3x^2 - 3yz$$

$$= \frac{3(x^2 - yz)}{(x^3 + y^3 + z^3 - 3xyz)}$$

$$\frac{\partial}{\partial y} = \frac{1}{(x^3 + y^3 + z^3 - 3xyz)} \times 3y^2 - 3xz$$

$$= \frac{3(y^2 - xz)}{(x^3 + y^3 + z^3 - 3xyz)}$$

$$\frac{\partial}{\partial z} = \frac{1}{(x^3 + y^3 + z^3 - 3xyz)} \times 3z^2 - 3xy$$

$$= \frac{3(z^2 - xy)}{(x^3 + y^3 + z^3 - 3xyz)}$$

now,

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) u = \left[\frac{3}{x^3 + y^3 + z^3 - 3xyz} (x^2 - yz) + \frac{3}{x^3 + y^3 + z^3 - 3xyz} (y^2 - xz) + \frac{3}{x^3 + y^3 + z^3 - 3xyz} (z^2 - xy) \right]$$

$$= \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{(x^3 + y^3 + z^3 - 3xyz)}$$

$$\frac{3(x^2+y^2+z^2-xy-yz-zx)}{(x+y+z)(x^2+y^2+z^2-xy-yz-zx)}$$

$$= \frac{3}{x+y+z}$$

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = \frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \left(\frac{3}{x+y+z} \right)$$

$$= \frac{\partial}{\partial x} \left(\frac{3}{x+y+z} \right) + \frac{\partial}{\partial y} \left(\frac{3}{x+y+z} \right) + \frac{\partial}{\partial z} \left(\frac{3}{x+y+z} \right)$$

$$= -3(x+y+z)^{-2} - 3(x+y+z)^{-2} - 3(x+y+z)^{-2}$$

$$= \frac{-9}{(x+y+z)^2} =$$

Q5) $u = \tan^{-1} \left(\frac{x^3+y^3}{x-y} \right)$ Prove that

$$i) x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$$

$$ii) x^2 \frac{\partial^2 u}{\partial x^2} + y^2 \frac{\partial^2 u}{\partial y^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} = 2 \cos 3u \sin u$$

$$\Rightarrow \tan u = z = \frac{x^3+y^3}{x-y}$$

$$z = \frac{x^3 (1 + \frac{y^3}{x^3})}{x (1 - \frac{y}{x})}$$

$$z = \frac{x^2 (1 + \frac{y^3}{x^3})}{(1 - \frac{y}{x})}$$

So, $n = 2$.

① by deduction n

$$x \frac{\partial y}{\partial x} + y \frac{\partial y}{\partial y} = n \frac{f(u)}{f'(u)}$$

$$x \frac{\partial y}{\partial x} + y \frac{\partial y}{\partial y} = 2 \cdot \frac{\tan u}{\sec^2 u}$$

$$x \frac{\partial y}{\partial x} + y \frac{\partial y}{\partial y} = 2 + \frac{\sin u}{\cos u} \cdot \cos u$$

$$x \frac{\partial y}{\partial x} + y \frac{\partial y}{\partial y} = 2 \sin u \sin 2u$$

$$(1) \quad x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u) [g'(u) - 1]$$

$$\begin{aligned} &= \sin 2u [2 \cos 2u - 1] \\ &= 2 \sin 2u \cos 2u - \sin 2u \\ &= \sin 4u - \sin 2u \end{aligned}$$

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2 \cos 3u \sin u.$$

Tute - IV

Q1) If $y_1 = \frac{x_2 x_3}{x_1}$, $y_2 = \frac{x_3 x_1}{x_2}$, $y_3 = \frac{x_1 x_2}{x_3}$ show

that the Jacobian of $\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = 1$.

$$\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = \begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \frac{\partial y_1}{\partial x_3} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \frac{\partial y_2}{\partial x_3} \\ \frac{\partial y_3}{\partial x_1} & \frac{\partial y_3}{\partial x_2} & \frac{\partial y_3}{\partial x_3} \end{vmatrix}$$

$$\frac{\partial y_1}{\partial x_1} = -\frac{x_2 x_3}{x_1^2}, \quad \frac{\partial y_1}{\partial x_2} = \frac{x_3}{x_1}, \quad \frac{\partial y_1}{\partial x_3} = \frac{x_2}{x_1}$$

$$\frac{\partial y_2}{\partial x_1} = \frac{x_3}{x_2}, \quad \frac{\partial y_2}{\partial x_2} = -\frac{x_3 x_1}{x_2^2}, \quad \frac{\partial y_2}{\partial x_3} = \frac{x_1}{x_2}$$

$$\frac{\partial y_3}{\partial x_1} = \frac{x_2}{x_3}, \quad \frac{\partial y_3}{\partial x_2} = \frac{x_1}{x_3}, \quad \frac{\partial y_3}{\partial x_3} = -\frac{x_1 x_2}{x_3^2}$$

$$= \begin{vmatrix} -\frac{x_2 x_3}{x_1^2} & \frac{x_3}{x_1} & \frac{x_2}{x_1} \\ \frac{x_3}{x_2} & -\frac{x_3 x_1}{x_2^2} & \frac{x_1}{x_2} \\ \frac{x_2}{x_3} & \frac{x_1}{x_3} & -\frac{x_1 x_2}{x_3^2} \end{vmatrix}$$

$$\frac{1}{x_1^2 x_2^2 x_3^2} \begin{vmatrix} -x_2 x_3 & x_3 x_1 & x_1 x_2 \\ x_3 x_2 & -x_1 x_3 & x_1 x_2 \\ x_2 x_3 & x_1 x_3 & -x_1 x_2 \end{vmatrix}$$

$$\frac{(x_2 x_3)(x_3 x_1)(x_1 x_2)}{x_1^2 x_2^2 x_3^2} \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= -1(0) - 1(-2) + 1(2) = 2 + 2 = 4$$

Q2.) find the value of Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$

$u = (x^2 - y^2), v = 2xy, x = r \cos \theta, y = r \sin \theta$

$x^2 + y^2 = r^2$

$\Rightarrow \frac{\partial(u,v)}{\partial(x,y)} = \frac{\partial(u,v)}{\partial(r,\theta)}$

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix}$$

$\Rightarrow 4(x^2 + y^2)$

Q3.)

IF

Sho

x

$\frac{\partial(u,v)}{\partial(x,y)}$

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$= \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$\Rightarrow r \cos^2 \theta + r \sin^2 \theta$$

$$\Rightarrow r (\cos^2 \theta + \sin^2 \theta) = r$$

now,

$$\frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(r, \theta)} \Rightarrow 4(r^2 + r^2) \times r \rightarrow r^2$$

$$\Rightarrow 4r^3 = \frac{\partial(u, v)}{\partial(r, \theta)}$$

Q3.) If $x + y + z = u$, $y + z = uv$, $z = uvw$
Show that $\frac{\partial(x, y, z)}{\partial(u, v, w)} = u^2 v$

$$x = u - y + z, \quad y = uv - z, \quad z = uvw$$

$$= u - uv, \quad y = uv - uvw$$

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

$$\frac{\partial x}{\partial u} = -\frac{(1-v)}{v}, \frac{\partial x}{\partial v} = -4, \frac{\partial x}{\partial w} = 0$$

$$\frac{\partial y}{\partial u} = v(1-w), \frac{\partial y}{\partial v} = u(1-w), \frac{\partial y}{\partial w} = -uv$$

$$\frac{\partial z}{\partial u} = vw, \frac{\partial z}{\partial v} = uw, \frac{\partial z}{\partial w} = uv$$

$$= \begin{vmatrix} (1-v) & -4 & 0 \\ v(1-w) & u(1-w) & -uv \\ vw & uw & uv \end{vmatrix}$$

$$(1-v) [u(1-w)uv + u^2wv] + 4 [uv(v(1-w)) + uv \times vw]$$

$$(1-v) [u^2v - u^2vw + u^2wv] + 4 [uv^2 - uv^2w + uv^2w]$$

$$(1-v) [u^2v] + u^2v^2$$

$$u^2v + uv^2 - u^2v^2 - uv^3$$

$$u^2v [1 - \frac{v}{u} - v]$$

$$u^2v - u^2v^2 + 4^2v^2$$

$$= u^2v$$

Q4) expand $e^x \cos y$ as (424)

$$f(x, y) = e^x \cos y$$

$$f(x, y) = e^x \cos y \quad e/\sqrt{2}$$

$$f_x(x, y) = -e^x \cos y \quad e/\sqrt{2}$$

$$f_y(x, y) = -e^x \sin y \quad -e/\sqrt{2}$$

$$f_{xx}(x, y) = e^x \cos y \quad e/\sqrt{2}$$

$$f_{yy}(x, y) = -e^x \cos y \quad -e/\sqrt{2}$$

$$f_{yx}(x, y) = -e^x \sin y \quad -e/\sqrt{2}$$

$$f_{xxx}(x, y) = e^x \cos y \quad e/\sqrt{2}$$

$$f_{yyy}(x, y) = -e^x \sin y \quad -e/\sqrt{2}$$

$$f_{xxy}(x, y) = -e^x \sin y \quad -e/\sqrt{2}$$

$$f_{xyy}(x, y) = -e^x \cos y \quad -e/\sqrt{2}$$

$$f(x, y) = \frac{e}{\sqrt{2}} + \frac{1}{\sqrt{2}} \left(\frac{x^2 e}{\sqrt{2}} - \frac{y^2 e}{\sqrt{2}} \right) + \frac{1}{\sqrt{2}} \left(\frac{x^2 e}{\sqrt{2}} - \frac{y^2 e}{\sqrt{2}} \right)$$

$$- 2xye \bigg) + \frac{1}{\sqrt{3}} \left(\frac{x^3 e}{\sqrt{2}} + \frac{y^3 e}{\sqrt{2}} - \frac{3x^2 y e}{\sqrt{2}} \right)$$

$$- \frac{3xy^2 e}{\sqrt{2}} \bigg) + \dots$$

$$= \frac{e}{2} \left[1 + x - y + \frac{1}{2} (x^2 - y^2 - 2xy) + \frac{1}{6} (x^3 + y^3 - 3x^2 y - 3xy^2) + \dots \right]$$

5)

$$f(x, y) = e^x \sin y$$

		$x=0$	$y=0$
$f(x, y)$	$e^x \sin y$	0	0
$f_x(x, y)$	$e^x \sin y$	0	0
$f_y(x, y)$	$e^x \cos y$	1	1
$f_{xx}(x, y)$	$e^x \sin y$	0	0
$f_{yy}(x, y)$	$-e^x \sin y$	0	0
$f_{yx}(x, y)$	$e^x \cos y$	1	1
$f_{xxx}(x, y)$	$e^x \sin y$	0	0
$f_{yyy}(x, y)$	$-e^x \cos y$	-1	-1
$f_{xyy}(x, y)$	$e^x \cos y$	1	1
$f_{xyx}(x, y)$	$-e^x \sin y$	0	0

By Taylor's theorem

$$f(x, y) \approx 0 + [x(0) + y(1)]$$

$$+ \frac{1}{2!} [x^2(0) + y^2(0) + 2xy(1)] + \frac{1}{3!} [x^3(0) + y^3(0) + 3x^2y(1) + 3xy^2(0)]$$

$$= y + xy - \frac{y^3}{6} + 3x^2y + \dots$$

$$= y + xy + 3x^2y - \frac{y^3}{6} + \dots$$